Adam Gordon-Fennell, Ph.D.

The University of Washington
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EDUCATION

University of Texas at Austin, Ph.D. **2014 –2019**

- Program: Institute for Neuroscience

University of California, Davis, B.S. **2009 –2012**

- Major: Psychology with an emphasis in Biology

FUTURE POSITION

Rutgers University **September 2026**

- Assistant Professor (tenure-track), Department of Psychology.

RESEARCH EXPERIENCE

University of Washington **2020 – Current**

Postdoctoral Research Fellow; ***Advisor: Garret Stuber***

Hypothalamic and dopamine circuits of operant and consummatory behaviors in mice. Ongoing projects include determining the functional coupling between lateral hypothalamic subpopulations and dopamine release and describing the striatum-wide spatiotemporal dynamics of dopamine release during consumption and reinforcement behaviors.

- Designed and validated an open-source head-fixed behavioral system (OHRBETS) for measuring operant behaviors including instrumental conditioning for sucrose, multiple solution brief access taste task, intra-cranial self-stimulation, and real-time place preference.
- Worked with over 10 labs to implement OHRBETS or design novel approaches for conducting behavioral experiments in head-fixed mice.
- Determined the tandem signaling dynamics of lateral hypothalamus GABA and glutamate neurons during consummatory behavior using dual-color fiber photometry.
- Determined the functional relationship between lateral hypothalamus GABA and glutamate neurons and dopamine release across the striatum by establishing and using multi-site fiber photometry and optogenetics.
- Trained in 2-photon calcium imaging and holographic optogenetics using spatial light modulation (SLM) during head-fixed behavior.
- Trained in analysis to determine the relationship between neuronal dynamics and behavior including generalized linear models (GLM) and classification using machine learning models.

University of Texas at Austin **2014 – 2019**

Graduate Researcher; ***Advisor: Michela Marinelli***

Functional connectivity between the lateral preoptic area (LPO) and ventral tegmental area (VTA) neurons in rats. LPO and LPO→VTA regulation of reward behavior in rats.

- Determined the causal role of the LPO in reward behavior by setting up and utilizing *in-vivo* awake behaving optogenetics.
- Determined the functional connectivity between the LPO and VTA with *in-vivo* anesthetized electrophysiology recordings and optogenetic manipulation of the LPO and LPO→VTA pathway.
- Explored the role of the LPO in reinstating cocaine and sucrose seeking behavior.
- Recorded the LPO during reward behaviors by establishing and using fiber photometry.

University of California, Davis **2012 – 2014**

Junior Specialist; ***Advisor: Leah Krubitzer***

Posterior parietal lobe regulation of motor behaviors and neural activity in primary somatosensory cortex in non-human primates

- Trained three macaques to perform a reaching and grasping behavioral procedure.
- Assessed the effect of reversibly deactivating posterior parietal lobe on reaching and grasping using video coding and behavioral analysis.
- Assisted with *in-vivo* multi-unit recordings of somatosensory cortex and analysis.

PUBLICATIONS

2026 Hjort M.H., Garrett Z.Q., **Gordon A.**, Ancell E., Trzeciak T., Lu P., Bruchas M.R., Witten D.M., Steinmetz N.A., Stuber G.D. Neural Dynamics of a Prefrontal Cortex to Ventral Tegmental Area Circuit Drive Cognitive Flexibility by Accelerating Contingency Degradation. *Nature*. (DOI:)

2025 Volcko K.L., Gresch A., Benowitz B.M., Taghipourbibalan H., Visser M., Stuber G.D., Gordon-Fennell A., Patriarchi T., McCutcheon J.E. Histamine Dynamics During Ingestive Behavior Measured by the Novel Biosensor HisLightG. Journal of Neurochemistry (DIO: 10.1111/jnc.70142).

2024 Hjort M.M., Gowrishankar R., Tian L., **Gordon-Fennell A.**, Namboodiri V.M.K., Bruchas M.R., Stuber G.D. Microprisms enable enhanced throughput and resolution for longitudinal tracking of neuronal ensembles in deep brain structures. *Neurophotonics* (DOI: 10.1117/1.NPh.11.3.033407)

2023 Zhou C., **Gordon-Fennell A.**, Piantadosi S.C., Ji N., Smith S.L., Bruchas M.R., and Stuber G.D. Deep-brain optical recording of neural dynamics during behavior. *Neuron* (DOI: 10.1016/j.neuron .2023.09.006)

Gordon-Fennell A., Barbakh J.M., Utley M., Singh S, Bazzino P., Gowrishankar R., Bruchas M.R., Roitman M.F., Stuber G.D. An Open-Source Platform for Head-Fixed Operant and Consummatory Behavior. *eLife* (DOI: 10.7554/eLife.86183)

eLife assessment: This **important** study by Gordon-Fennell et al. presents a low-cost, open-source platform for measuring action elicitation and consummatory behavior in head-fixed animals. The authors present **exceptional** evidence that this platform allows animals to perform a truly voluntary action whilst their head is held still. The results have the potential to have a broad impact in the field as many labs start to move towards measuring head-fixed behavior effectively, although this is said with the caveat that such behavior will never be an ideal replication of naturalistic behavior [*bolding by author*].

2021 **Gordon-Fennell A.**, and Stuber G.D. Illuminating Subcortical GABAergic and Glutamatergic Circuits for Reward and Aversion. *Neuropharmacology* (DOI: 10.1016/j.neuropharm.2021.108725)

2020 **Gordon-Fennell A.**, Gordon-Fennell L., Desaivre S., Marinelli M. The Lateral Preoptic Area and Its Projection to the VTA Regulate VTA Activity and Drive Complex Reward Behaviors. *Frontiers in Systems Neuroscience* (DOI: 10.3389/fnsys.2020.581830)

Gordon-Fennell A., Will R., Ramachandra V., Gordon-Fennell L., Dominguez J., Zahm D., Marinelli M. The lateral preoptic area: a novel regulator of reward seeking and neuronal activity in the ventral tegmental area. *Frontiers in Neuroscience* (DOI: 10.3389/fnins.2019.01433)

2019 Pomrenze M.B., Giovanetti S.M., Maiya R., **Gordon A.G.**, Kreeger L.J., Messing R.O. Dissecting the Roles of GABA and Neuropeptides from Rat Central Amygdala CRF Neurons in Anxiety and Fear Learning. *Cell Reports* (DOI: 10.1016/j.celrep.2019.08.083)

2014 Goldring A.B., Cooke D.F., Baldwin M.K., Recanzone G.H., **Gordon A.G.,** Pan T., Simon S.I., Krubitzer L. Reversible deactivation of higher-order posterior parietal areas. II. Alterations in response properties of neurons in areas 1 and 2. *Journal of Neurophysiology* (DOI: 10.1152/jn.00141.2014)

For a full list of publications see https://scholar.google.com/citations?user=aVheI74AAAAJ&hl=en

PUBLICATIONS IN PROCESS

- **Gordon-Fennell A.**, Benowitz B.M., Barbakh J.M., Montequin I., Campuzano A., Stevenson H., Hjort M.M., Critz M., Ancell E., Witten D. M., Stuber G.D. *Working title:* The lateral hypothalamus regulates striatum wide dopamine dynamics during consumption. *In revisions at Neuron* (bioRxiv DOI: 10.1101/2025.04.09.648052).
- Doncheck E.M.*, **Gordon-Fennell A.***, Boquiren J., Manusky L.M., Baek J., Clarke R.E., Paniccia J.E., Grant R.I., Stuber G.D., Otis J.M. *Working title:* Fabrication and implementation of drug self-administration experiments in head-fixed mice. *In revisions at Nature Protocols.* (*authors contributed equally)
- Gowrishankar R., Hjort M.M., Elerding A.J., Shirley S.E., Tilburg J.V., Marcus D.J., Abrera K.A., Senthilkumar P., Sumarli D., Motovilov K., Lau V., **Gordon-Fennell A.**, Zhou Z.C., Dong C., Tian Lin., Stuber G.D., Bruchas M.R. *Working title:* Endogenous opioid dynamics in the dorsal striatum shape goal-directed behavior. *In revisions at Cell* (bioRxiv DOI: 10.1101/2024.05.20.595035).
- Loveless M.C., Bryan C., Tchumi C., **Gordon-Fennell A.**, Morton G., Soden M. *Working title:* Lateral hypothalamus neurotensin neurons regulate energy balance and encode consummatory reward.

GRANTS

2024 – 2029	NIDA **K99DA059612**
2022 – 2023	NIDA **F32DA054719;** $70,282
2022 – 2023	NIDA **P30DA048736**, Pilot Project; $44,844
2020 – 2022	NIDA **T32DA7278**; $24,324
2018 – 2019	NIAAA **T32AA007471**; $24,324
2018 – 2019	Waggoner Center Bruce Jones Fellowship, $16,667
2015 – 2018	NSF Graduate Research Fellow, **NSF GRFP DGE-1110007**; $138,000

HONORS AND FELLOWSHIPS

2025	Winter Conference on Brain Research - *Grand Outstanding Poster*
2024	Winter Conference on Brain Research - *Poster Finalist*
	Winter Conference on Brain Research - *Travel Fellow*
2017	Graduate Student Professional Development Award Travel Grant
2015 – 2017	Graduate Dean’s Prestigious Fellowship Supplement Fellow
2015	Graduate Student Professional Development Award travel grant
2012	Graduated from UC Davis with honors

ABSTRACTS AND PRESENTATIONS

Selected and Invited Oral Presentations

2026 **Gordon-Fennell A.** Dopamine signals across the anterior-posterior axis of the striatum represent unique components of consummatory behavior and causally shape consumption. Presented at Optogenetic Approaches to Understanding Neural Circuits and Behavior Gordon Research Confrence, Lucca, Italy. ***Selected speaker***

2025 **Gordon-Fennell A**. Opposing Hypothalamic Pathways Orchestrate Striatum-Wide Dopamine Dynamics During Consumption. Presented at University of Calgary and the Hotchkiss Brain Institute Seminar Series, Calgary, Canada. ***Invited speaker.***

Gordon-Fennell A. Beyond 1 fiber: Techniques for multi-site and multi-subject fiber photometry. Presented at University of Calgary and the Hotchkiss Brain Institute Fiber Photometry Workshop, Calgary, Canada. ***Invited speaker.***

Gordon-Fennell A. Dopamine signals across the anterior-posterior axis of the striatum represent unique components of consummatory behavior. Presented at CoNECTOME, Seattle, WA. ***Invited speaker.***

Gordon-Fennell A. Dopamine signals across the anterior-posterior axis of the striatum represent unique components of consummatory behavior. Presented at Conference on Stress, Pain, Emotion, Addiction, and Reward (SPEAR), Friday Harbor, WA. ***Selected speaker.***

2024 **Gordon-Fennell A.** The Lateral Hypothalamus Regulates Striatum-Wide Dopamine Dynamics During Consumption of Rewarding and Aversive Solutions. Presented at Optogenetic Approaches to Understanding Neural Circuits and Behavior Gordon Research Seminar, Lucca, Italy. ***Selected speaker.***

Gordon-Fennell A. The Tandem Role of Lateral Hypothalamic GABA and Glutamate Neurons in Regulating Striatal Dopamine Release and Motivated Behavior. Presented at the Kobe University - University of Washington Alliance Project Symposium, Kobe, Japan. ***Invited speaker.***

2023 **Gordon-Fennell A.** Dynamics of Lateral Hypothalamus GABA and Glutamate Neurons and Dopamine Release During Consummatory Behaviors. Presented at Catecholamines Gordon Research Seminar, Castelldefels, Spain. ***Selected speaker.***

2022 **Gordon-Fennell A**. Mesolimbic Dopamine Signatures During Consumption of Appetitive and Aversive Tastants in a Novel Head-Fixed Behavioral System. Presented virtually at University of California Los Angeles, Behavioral Neuroscience Journal Club, Los Angeles, CA. ***Invited speaker*.**

Gordon-Fennell A. A Novel Head-Fixed Behavioral System Reveals Simultaneous Dynamics of Lateral Hypothalamus GABA and Glutamate Cells During Operant and Consummatory Behaviors. Presented at Hypothalamus Gordon Research Conference, Ventura, CA. ***Selected speaker.***

2021 **Gordon-Fennell A**. An open-source head-fixed system for measuring affective valence and motivated behavior. Presented at Graduate Neuroscience Retreat, University of Washington, WA**. *Invited speaker.***

2016 **Gordon, A**. The lateral preoptic area regulates dopamine neuron activity and cocaine seeking. University of Texas at Austin Institute for Neuroscience graduate recruitment. UT Austin, TX**. *Invited speaker.***

2015 **Gordon A.** Properties and Connectivity of Central Amygdala CRF Neurons. University of Texas at Austin Institute for Neuroscience Retreat. The Retreat at Balcones Springs, Marble Falls, TX ***Invited speaker.***

Oral Presentations

2025 **Gordon-Fennell A**. Pitfalls and Progress in Head-Fixed Negative Reinforcement Design. Presented at The Head-Fixed Behavior Symposium, Seattle, WA.

Gordon-Fennell A. Dopamine signals across the anterior-posterior axis of the striatum represent unique components of consummatory behavior. Presented at Winter Conference on Brain Research, Tahoe, CA.

Gordon-Fennell A. A Beginner's Guide to Programming in R and RStudio. Presented at Winter Conference on Brain Research, Tahoe, CA.

2024 **Gordon-Fennell A.** The Tandem Role of Lateral Hypothalamic GABA and Glutamate Neurons in Regulating Striatal Dopamine Release and Motivated Behavior. Presented at Conference on Stress, Pain, Emotion, Addiction, and Reward (SPEAR), Kananaskis Centre: Barrier Lake Field Station, Canada.

Gordon-Fennell A. The Tandem Role of Lateral Hypothalamic GABA and Glutamate Neurons in Regulating Striatal Dopamine Release and Motivated Behavior. Presented at Winter Conference on Brain Research, Breckenridge, CO.

2023 **Gordon-Fennell A.** From Raw Data to Publication: Coding and Illustrator Techniques for High Quality Figures. Presented at NAPE Computational Meeting, University of Washington, WA.

2022 **Gordon-Fennell A**. Tidy TDT: a Case Study in Automated and Reproducible Analysis. Presented at NAPE Computational Meeting, University of Washington, WA.

2021 **Gordon-Fennell A**. A tidy pipeline for behavioral data formatting, wrangling, analysis, and plotting. Presented at NAPE Computation Meeting, University of Washington, WA.

Gordon-Fennell A. An open-source head-fixed system for measuring affective valence and motivated behavior. Presented at NAPE seminar series, University of Washington, WA.

2019 Morales M. (Chair), Barker D. (Co-Chair), **Gordon A.**, and O'Connor R. Diverse Roles of the Hypothalamus and Hypothalamic Circuits in Motivation and Psychopathology. Panel presentation at Winter Conference on Brain Research, Snowmass, CO.

2018 **Gordon A**. The mysterious function of the lateral preoptic. Presented at Dopamine Club, University of Texas at San Antonio, TX.

2017 **Gordon A**. The lateral preoptic area regulates VTA neuron activity and reward behaviors. Presented at Presented at Waggoner Center for Alcohol & Addiction Research seminar, UT Austin, TX.

2016 **Gordon A**. The Lateral Preoptic Area: A Novel Regulator of VTA Activity and Cocaine Seeking. Presented at WCAAR Advance, UT Austin, TX.

2015 **Gordon, A**. Effect of Food Restriction Stress on Dopaminergic Transmission, Cocaine Taking, and Cocaine Seeking. Presented at Pharmacology and Toxicology Seminar Series. UT Austin, TX.

Gordon A. Properties and Connectivity of Central Amygdala CRF Neurons. Presented at Waggoner Center for Alcohol & Addiction Research seminar, UT Austin, TX.

Gordon A. Effect of Food Restriction Stress on Dopaminergic Transmission, Cocaine Taking, and Cocaine Seeking. Presented at Waggoner Center for Alcohol & Addiction Research seminar, UT Austin, TX.

Chaired Pannels

2025 **Gordon-Fennell A. (Chair)**, Faget L., Calipari E.S., and Lammel S. Carving New Trails in Dopamine Signaling: Investigating Dopamine Subpopulations and Their Role in Reward, Aversion, and Motivation Behaviors. Winter Confrence on Brain Research, Tahoe, CA.

2024 **Gordon-Fennell A. (Chair)**, Barbano F. (Co-chair), Eban-Rothschild A., and Navratilova E. Regulation of Diverse Motivated Behaviors by the Hypothalamus. Presented at Winter Conference on Brain Research, Breckenridge, CO.

Symposiums Organized

2025 **Lead Organizer,** *The Head-Fixed Behavior Symposium*, University of Washington. Half-day symposium to foster knowledge exchange on head-fixed behavioral paradigms in neuroscience, highlight methodological innovations, and promote collaboration among labs developing and using these approaches.

Poster Presentations

2026 **Gordon-Fennell A.**, Benowitz, B.M., Lu P., Barbakh J.M., Montequin I., Campuzano A., Ancell E., Hjort M.M., Witten D., and Stuber G.D. Dopamine signals across the anterior-posterior axis of the striatum represent unique components of consummatory behavior and causally shape consumption. Presented at Optogenetic Approaches to Understanding Neural Circuits and Behavior Gordon Research Confrence, Lucca, Italy.

Gordon-Fennell A., Lu E., Campuzano A., Stuber G.D. Causal roles of dopamine across striatal subregions in consumption behavior. Presented at Conference on Stress, Pain, Emotion, Addiction, and Reward (SPEAR), Kananaskis Centre: Barrier Lake Field Station, Canada.

Gordon-Fennell A., Benowitz, B.M., Barbakh J.M., Montequin I., Campuzano A., Stevenson H., Lu E., Hjort M.M., Ancell E., Critz, M., Witten D., and Stuber G.D. Lateral hypothalamic GABA and glutamate neurons cooperatively regulate striatum-wide dopamine release during consummatory behavior. Presented at Winter Confrence on Brain Research, Big Sky, MT.

2025 **Gordon-Fennell A.**, Benowitz, B.M., Lu E., Montequin I., Campuzano A., Stevenson H., Hjort M.M., Benowitz B.M., and Stuber G.D. Dopamine signals across the anterior-posterior axis of the striatum represent unique components of consummatory behavior. Presented at Winter Conference on Brain Research, Tahoe, CA. ***Grand Outstanding Poster.***

Gordon-Fennell A., Benowitz, B.M., Lu E., Montequin I., Campuzano A., Stevenson H., Hjort M.M., Benowitz B.M., and Stuber G.D. Dopamine signals across the anterior-posterior axis of the striatum represent unique components of consummatory behavior. Presented at Anesthesiology & Pain Medicine Academic Evening, Seattle, WA.

Gordon-Fennell A., Benowitz, B.M., Lu E., Montequin I., Campuzano A., Stevenson H., Hjort M.M., Benowitz B.M., and Stuber G.D. Dopamine signals across the anterior-posterior axis of the striatum represent unique components of consummatory behavior. Presented at Society for Neuroscience, San Diego, CA.

2024 **Gordon-Fennell A.**, Hjort M.M., Benowitz B.M., and Stuber G.D. The lateral hypothalamus regulates striatum wide dopamine dynamics during consumption of rewarding and aversive solutions. Presented at Optogenetic Approaches to Understanding Neural Circuits and Behavior Gordon Research Seminar, Lucca, Italy.

Gordon-Fennell A., Barbakh J.M., Benowitz B.M., and Stuber G.D. The Tandem Role of Lateral Hypothalamic GABA and Glutamate Neurons in Regulating Striatal Dopamine Release and Motivated Behavior. Presented at Winter Conference on Brain Research, Breckenridge, CO. ***Poster Award Finalist.***

2023 **Gordon-Fennell A.**, Barbakh J.M., Benowitz B.M., and Stuber G.D. Dynamics of Lateral Hypothalamus GABA and Glutamate Neurons and Dopamine Release During Consummatory Behaviors. Presented at Catecholamines Gordon Research Conference, Castelldefels, Spain.

2022 **Gordon-Fennell A**., Gowrishankar R., Bruchas M.R., and Stuber G.D. A Novel Head-Fixed Behavioral System Reveals Simultaneous Dynamics of Lateral Hypothalamus GABA and Glutamate Cells During Operant and Consummatory Behaviors. Presented at Hypothalamus Gordon Research Conference, Ventura, CA.

2018 **Gordon A.**, Fennell F., Fang M., Marinelli M. A novel method for quantifying regional distribution of neural manipulations relative to a reference atlas. Presented at Neuroscience, San Diego, CA.

Gordon A., Ramachandra V.S., Mittal N., Duvauchelle C., and Marinelli M. Optogenetic activation of the lateral preoptic area excites dopamine neurons, supports self-stimulation, and elicits “positive affect” ultrasonic vocalizations. Presented at WCAAR Advance, UT Austin, TX.

2017 **Gordon A.,** Ramachandra V.S., Mittal N., Duvauchelle C., and Marinelli M. Optogenetic activation of the lateral preoptic area excites dopamine neurons, supports self-stimulation, and elicits “positive affect” ultrasonic vocalizations. Presented at Neuroscience, Washington, DC.

2015 **Gordon, A.,** Ramachandra V.S., Fitzgerald A., and Marinelli M. Food restriction stress enhances cocaine seeking and VTA dopamine neuron activity. Presented at Neuroscience, Chicago, IL.

Gordon, A., Ramachandra V.S., and Marinelli M. Food restriction stress enhances cocaine seeking and VTA dopamine neuron activity. Presented at Pharmacy Research Day, UT Austin, TX.

Gordon, A., Ramachandra V.S., and Marinelli M. Food restriction stress enhances cocaine seeking and VTA dopamine neuron activity. Presented at WCAAR Advance, UT Austin, TX.

DIVERSITY, EQUITY, AND INCLUSION

2023 – 2025 Enhancing Neuroscience Diversity through Undergraduate Research Education Experiences (ENDURE) ***Mentor***

2023 – current Letters to a Pre-Scientist

2022 Project SHORT ***Mentor***

2022 Empowering Prevention & Inclusive Communities (EPIC) ***Training***

PROFESSIONAL DEVELOPMENT AND ACADEMIC SERVICE

2025 Cold Spring Harbor Laboratory, Imaging Structure and Function in the Nervous System - Student

2024 **Gordon-Fennell A.** and Piantadosi S. S. Making Your Research Project Sparkle. Presented at the NAPE Center Graduate Student Professional Development Seminar Series, University of Washington, WA.

2023 **Gordon-Fennell A.,** Gowrishankar R.**,** and Coffey K. R. Pursuing an Academic Career. Presented at the NAPE Center Graduate Student Professional Development Seminar Series, University of Washington, WA

COMMUNITY OUTREACH AND SERVICE

Addiction Education Outreach

2015 – 2019 **Gordon A.**, Cofresí R., and Warden A. Neuroscience of Addiction Youth Outreach. Public schools, Travis County, TX.

Designed and operated drug education youth outreach program aimed at *teaching middle school and high school students (primarily low SES)* about the neuroscience of drugs and addiction. **5 schools, over 1200 students across 62 class periods.**

2018 Lifeworks, Austin, TX. Presented information on the neurobiology of drug addiction to homeless youth hosted by Lifeworks.

Youth Substance Abuse Prevention Coalition (YSAPC), Austin, TX. Presented information on the neurobiology of drug addiction to caregivers and members of YASAPC

Neuroscience Outreach

2018 UT Brainstorms: A Conversation on the Brain. Panel discussion on neurobiology and treatment of addiction. UT Austin, TX. ***Pannel moderator***

Thinkery: All Things Human, Austin, TX. Taught the lay public about the brain and its ability to create human experience and behavior.

Academic Service

2024 ***Reviewer*** - Addictions, drug & Alcohol Institute (ADAI) small grants.

2017 – 2018 Neuroscience Graduate Student Association Secretary.

TEACHING EXPERIENCE

Guest Lecturer

2024 Rats as a Model System for Substance Use Disorder. Presented at NEUSCI 450 undergraduate neuroscience class. University of Washington, WA.

2023 Neuronal Circuits of Sodium Appetite. Presented at NEURO 511 graduate neuroscience class. University of Washington, WA

2021 Emerging Technologies for Behavioral Neuroscience. Presented at NEURO 511 graduate neuroscience class. University of Washington, WA

Workshops

NAPE Calcium Imaging Workshop, University of Washington (annual, 2021 – 2024*)*

- Led an annual workshop (4h) on open-source hardware and software for conducting head-fixed two-photon experiments with a focus on Arduino.
- Co-led a workshop (4h) with Carrie Stine on methods for conducting and analyzing fiber photometry experiments.
- Reviewed behavioral considerations for one-photon and two-photon experiments with a focus on challenges associated with head-fixed experiments.
- Recruited participants from outside universities.

Teaching Assistant (Spring 2019)

Course on Neurobiology of Addiction (94 students), Department of Neuroscience, University of Texas at Austin

- Wrote and graded multiple choice and open-ended exams.
- Taught review sessions prior to exams.
- Created and presented a lecture on a scientific paper suitable for undergraduates.

Neuroscience Undergraduate Reading Program (2017 – 2019)

Department of Neuroscience, University of Texas at Austin

- Mentored an undergraduate student each semester on reading scientific papers in a field of interest.
- Taught students how to approach reading primary research articles, synthesize information across papers, and give an oral presentation.

SCIENTIFIC RESOURCE MANAGEMENT

UW NAPE 3d Printing Resources

- Built 3d printed resource center for NAPE with Madelyn Hjort, including purchasing and management of a Form3 SLA resin printer and an Ultimaker S3 filament printer.
- Managed training, scheduling, and maintenance of printers.

Mentorship

Stuber lab, University of Washington

Year	Name	Position	Current Position
2025 – current	Elena Lu	Research Technician	N/A
2024	Alex Garcia	Rotation Ph.D.	Ph.D. student, Stuber lab, University of Washington
2024	Madelyn Critz	Rotation Ph.D.	Ph.D. student, Land & Bruchas labs, University of Washington
2023 – 2026	Anthony Campuzano	Undergraduate Researcher (**ENDURE**)	N/A
2022 – 2025	Isabella Montequin	Undergraduate Researcher	N/A
2022	Sri Varshitha Pinnaka	Undergraduate Researcher	Undergraduate Researcher, Bruchas Lab, University of Washington
2021	Kaylin Elinof	Rotation Ph.D.	Ph.D. student, Land & Bruchas, University of Washington
2021	Jovana Navarrete	Rotation Ph.D.	Ph.D. student, Golden lab, University of Washington
2021 – 2025	Hannah Stevenson	Research Technician	N/A
2022 – 2024	Barbara Benowitz	Research Technician	Ph.D. student, Medical University of South Carolina
2021 – 2023	Shreya Singh	Undergraduate Researcher	Applying for medical school
2021 – 2023	Joumana Barbakh	Undergraduate Researcher (**NAPE Undergraduate Research Program**)	Accepted to medical school at the University of Washington
2020	Jessica Jones	Rotation Ph.D.	Ph.D. student, Tuthill lab, University of Washington
2020 – 2021	Taylor Hobbs	Research Technician	Ph.D. student, Gaunt lab, University of Pittsburgh
2020 – 2021	MacKenzie Utley	Undergraduate Researcher (**NAPE Undergraduate Research Program**)	Senior Clinical Research Associate; Applying for medical school

Marinelli lab, University of Texas Austin

Year	Name	Position	Current Position
2019	Kevin Sattler	Research Technician	Ph.D. candidate in Neuroscience, Zelikowsky lab, University of Utah
2019	Adrian Guanawan	Research Technician	
2018 – 2019	Lydia Fennell	Research Technician	Ph.D. candidate in Neuroscience, Phillips lab, University of Washington
2017 – 2019	Mary Fang	Undergraduate Researcher	General Surgery Resident Physician, Baylor College of Medicine
2017 – 2018	Kristina Zittel	Undergraduate Researcher	Climate Project Manager at U.S. Department of State, Georgetown University
2017	Alfredo Arellano	Undergraduate Researcher	
2016	Emma Brockway	Rotation Ph.D. Student	
2015	Philip Lambeth	Rotation Ph.D. Student	Postdoc, Margolis Lab, University of California San Francisco

REFERENCES

- **Dr. Garret Stuber**, Depts. of Anesthesiology and Pharmacology, UW (gstuber@uw.edu).
- **Dr. Michael Bruchas**, Depts. of Anesthesiology and Pharmacology, UW (mbruchas@uw.edu)
- **Dr. Michela Marinelli**, Dept. of Neurology, UT Austin (Micky.marinelli@austin.utexas.edu)
- **Dr. Mitchell Roitman**, Dept. of Psychology, UIC (mroitman@uic.edu)